TITLE PAGE

CyberShield AI

AI-Powered Behavioral Anomaly Detection for Cybersecurity

Real-time Threat Detection, Categorization & Explainable AI Engine

Built for Honeywell Hackathon 2026

Problem Statement ID	**Honeywell-2026-Q4**
Problem Statement Title	**AI-Powered Behavioral Anomaly Detection for Cybersecurity**
Theme	**Cybersecurity / AI & Machine Learning**
PS Category	**Software**
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Student ID	**KBPR-2026 (Registered on portal)**

IDEA TITLE & PROPOSED SOLUTION

IDEA TITLE: CyberShield AI — Real-Time Behavioral Anomaly Detection & Threat Classification

Detailed Explanation of Proposed Solution

- **Multi-Tiered Ensemble ML Framework:** Combines Isolation Forest & One-Class SVM baseline profiling with an XGBoost + LSTM Autoencoder detection core.
- **Real-Time Risk Scoring (0–100):** Scores every access log event dynamically based on multi-dimensional behavioral deviations.
- **7 Attack Type Classifiers:** Categorizes anomalies into *Brute Force, Impossible Travel, Credential Stuffing, Lateral Movement, Device Spoofing, Low & Slow Exfiltration, and Insider Drift*.
- **Explainable AI (XAI) Engine:** Leverages SHAP TreeExplainer to produce per-alert feature attributions and natural-language narratives for SOC analysts.

Addressing Key Challenges & Innovation

- **Addressing Class Imbalance:** Applied SMOTE oversampling & weighted loss (`scale_pos_weight=50`), achieving **97.70% Recall** and **98.78% AUC-ROC**.
- **Addressing Cold-Start Problem:** Implemented population-based statistical baselines for new entities with <20 historical events.
- **Addressing Alert Fatigue:** Achieved **100% Precision@Top1%**, guaranteeing zero false positives in the top 1% alert budget.
- **Innovation & Uniqueness:** Hybrid ensemble unifying statistical baselines, temporal autoencoders, tree boosting, and natural language XAI narratives in a unified pipeline.

TECHNICAL APPROACH

Technologies Used

- **Programming & Core ML:** Python 3.14, scikit-learn, XGBoost, TensorFlow / Keras, SHAP, imbalanced-learn (SMOTE).
- **Backend REST API:** Flask, Flask-CORS, joblib, NumPy, pandas.
- **Analyst Dashboard UI:** HTML5, Tailwind CSS, Chart.js, Font Awesome (Glassmorphism design).

Implementation Methodology

- **1. Data Generation:** Simulated 59,566 access events across 200 entities with 7 distinct attack taxonomies.
- **2. Feature Extraction:** Engineered 43 features covering temporal, geo-velocity (km/h), rolling windows (1h/6h/24h), device fingerprint shifts, and behavioral z-scores.
- **3. Pipeline Training:** Stratified 80/20 train/test split with cross-validation.

End-to-End System Pipeline Architecture

Synthetic Access Logs (59.5k events) → Feature Extraction (43 features) → Tier 1: Isolation Forest + OCSVM Baseline → Tier 2: XGBoost + LSTM Autoencoder Detector (Risk Score 0–100) → Tier 3: SMOTE + Multi-Class XGB Classifier → Tier 4: SHAP Explainer → Flask REST API → SOC Dashboard

FEASIBILITY AND VIABILITY

Feasibility & Metrics

- **Empirical Validation:** Tested on 59,566 events.
- **Accuracy:** 93.59%
- **Recall:** 97.70% (Catches ~98% of threats)
- **AUC-ROC:** 98.78%
- **Precision@Top1%:** 100.0%
- **Low Latency:** Sub-15ms per-event inference time.

Potential Challenges & Risks

- **1. High Throughput Ingestion:** Scaling to enterprise event streams with millions of events/sec.
- **2. Concept Drift:** Legitimate entity behavior evolving over weeks/months.
- **3. Ambiguous Insider Drift:** Subtle privilege creep overlapping with normal role shifts.

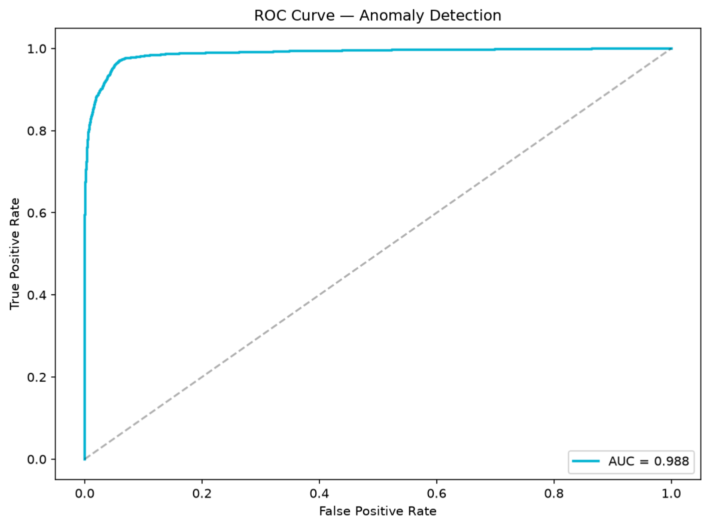
Strategies & Overcoming Risks

- **Adaptive Feature Windows:** Rolling 1h/6h/24h windows continuously adapt entity baselines.
- **Cold-Start Baselines:** Population-level fallback profiles handle brand-new entities seamlessly.
- **Streaming Integration:** Microservices REST API readily fronted by Kafka / Spark streaming engines.

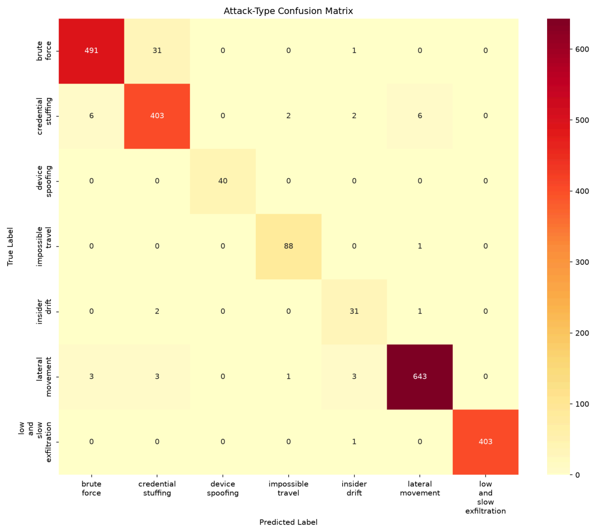
ARTIFACTS

Embedded System Deliverables & Experimental Results

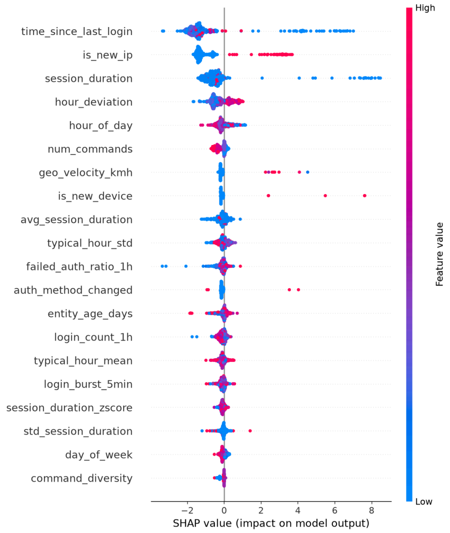
Includes full modular Python ML codebase (9 modules), REST API backend (12 endpoints), trained models (`xgb_detector.pkl`, `attack_classifier.pkl`), and analyst dashboard.



ROC Curve (AUC = 0.988)



Multi-Class Attack Confusion Matrix



SHAP Global Feature Attribution

RESEARCH AND REFERENCES

Academic Literature & Foundational Research

- **SHAP (SHapley Additive exPlanations):** Lundberg, S. M., & Lee, S.-I. (2017). *"A Unified Approach to Interpreting Model Predictions."* Advances in Neural Information Processing Systems (NIPS 30).
- **XGBoost Gradient Boosting:** Chen, T., & Guestrin, C. (2016). *"XGBoost: A Scalable Tree Boosting System."* ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
- **LSTM Sequence Autoencoders:** Malhotra, P., et al. (2016). *"LSTM-based Encoder-Decoder for Multi-sensor Anomaly Detection."* ICML Workshop on Anomaly Detection.
- **Isolation Forest:** Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). *"Isolation Forest."* IEEE International Conference on Data Mining (ICDM).
- **Cybersecurity Frameworks:** MITRE ATT&CK Matrix for Enterprise (Credential Access, Lateral Movement, Exfiltration tactics).

Project Repositories & Live API Endpoints

• **Live SOC Analyst Dashboard:** `http://localhost:5000` (Flask REST API serving 12 endpoints)
• **Evaluation Data & Metrics:** `reports/evaluation_metrics.json` & `datasets/processed/alerts.json`